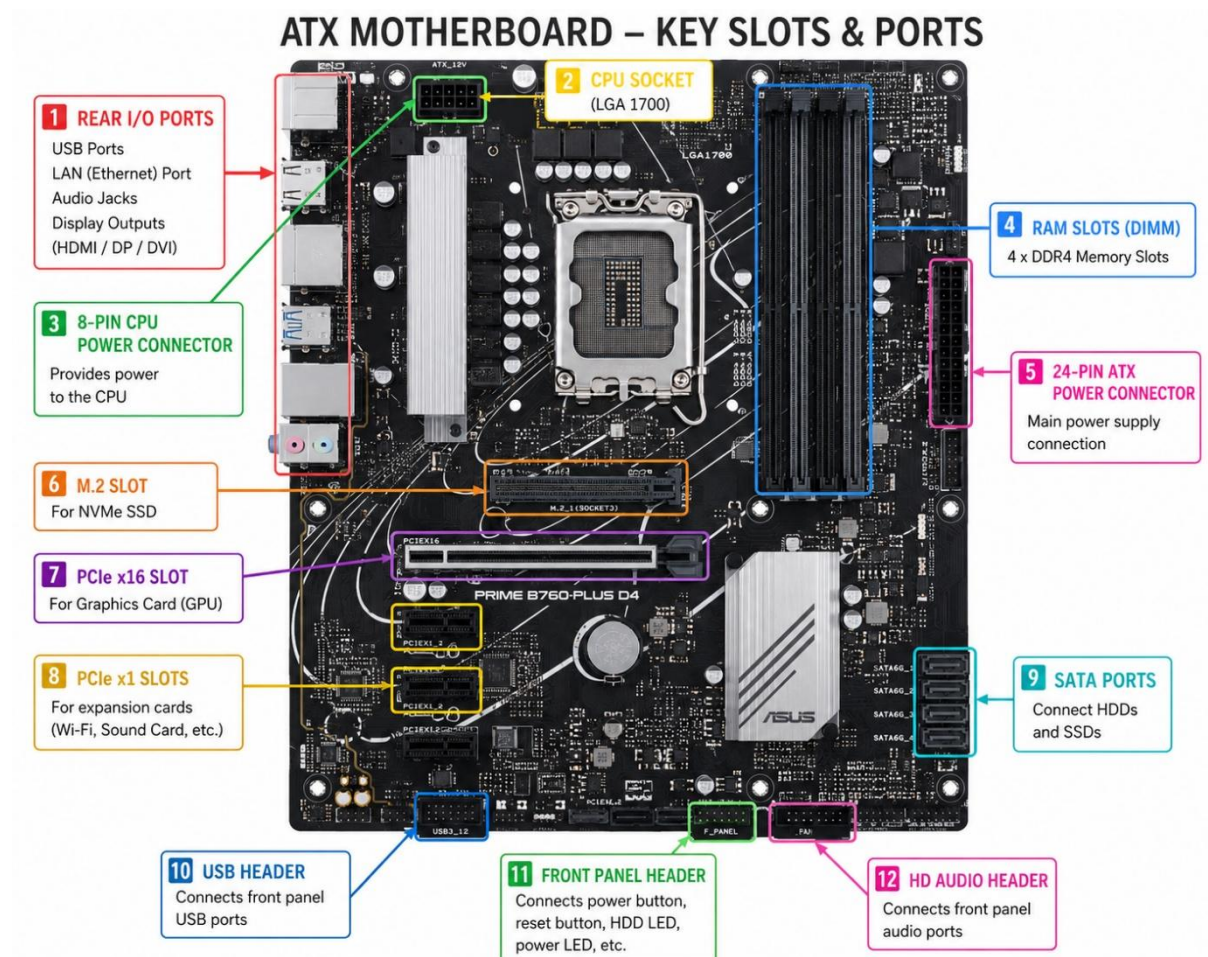


1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



Components of the ATX Motherboard Diagram

1. **Rear I/O Ports** – Used to connect external devices such as USB devices, monitor, speakers, and network cables.
2. **CPU Socket** – The slot where the Central Processing Unit (CPU) is installed.
3. **8-Pin CPU Power Connector** – Supplies power directly to the CPU.
4. **RAM Slots (DIMM Slots)** – Used to install memory modules (RAM).
5. **24-Pin ATX Power Connector** – Main power connector that supplies power to the motherboard.
6. **M.2 Slot** – Used for installing high-speed M.2 NVMe SSDs.
7. **PCIe x16 Slot** – Primarily used for installing a graphics card (GPU).
8. **PCIe x1 Slots** – Used for expansion cards such as Wi-Fi cards, sound cards, and network cards.
9. **SATA Ports** – Used to connect HDDs, SSDs, and optical drives.
10. **USB Header** – Connects the front-panel USB ports of the computer case to the motherboard.
11. **Front Panel Header** – Connects the power button, reset button, power LED, and HDD LED.
12. **HD Audio Header** – Connects the front-panel audio ports (headphone and microphone jacks) to the motherboard.

2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

CPU Socket Type	Description	Compatible CPU Model
LGA1151	Intel socket used for 6th, 7th, 8th, and 9th Generation Core processors.	Intel Core i7-9700K
AM4	AMD socket that supports Ryzen processors and Athlon CPUs.	AMD Ryzen 5 3600
LGA1200	Intel socket designed for 10th and 11th Generation Core processors.	Intel Core i5-10400
LGA1700	Intel socket used for 12th, 13th, and 14th Generation Core processors.	Intel Core i5-12400
sTRX4	AMD socket designed for Ryzen Threadripper processors.	AMD Ryzen Threadripper 3960X
TR4	AMD socket used for first and second generation Threadripper processors.	AMD Ryzen Threadripper 2950X

3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

CPU (Central Processing Unit):

The CPU is the main processor of the computer and is known as the brain of the system. It executes instructions, performs calculations, and processes data required by applications and hardware devices.

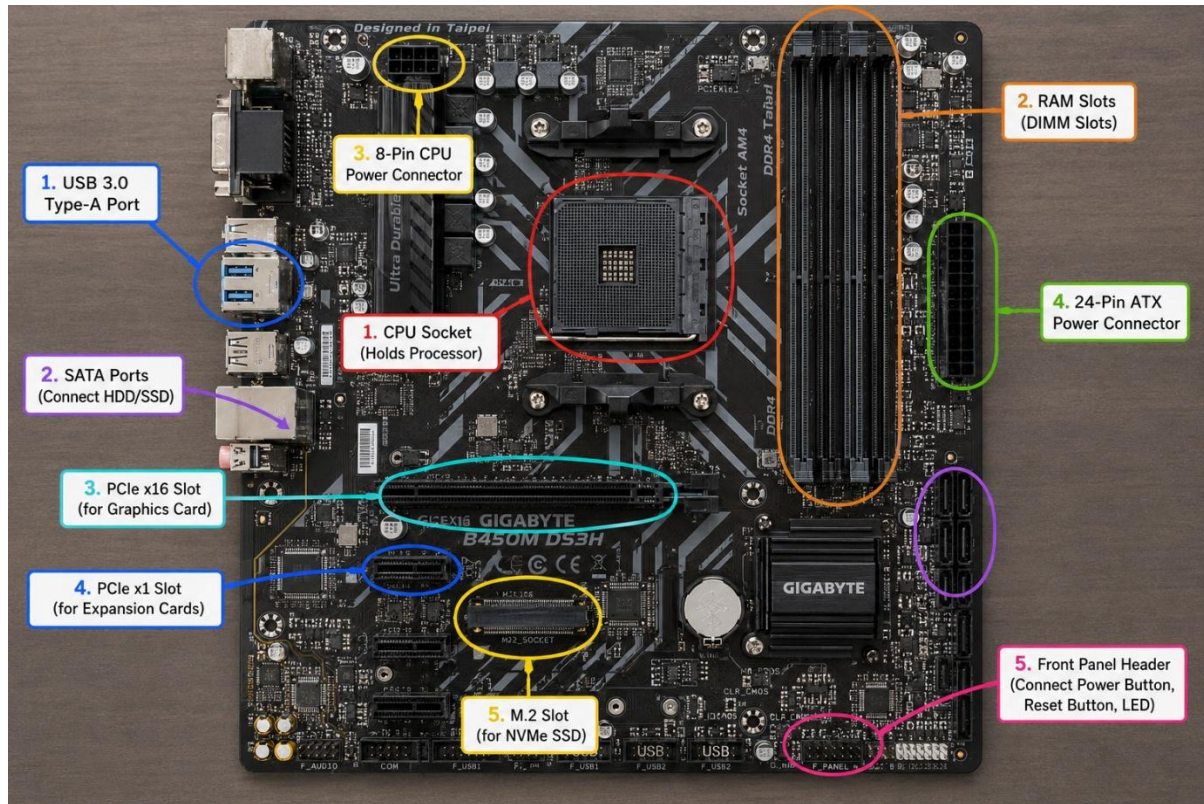
Chipset:

The chipset is a set of integrated circuits on the motherboard that controls communication between the CPU, RAM, storage devices, graphics card, and other peripherals. It helps different hardware components work together efficiently.

How They Work Together When Opening Spotify

1. The user clicks the Spotify icon.
2. The CPU receives the command to open the application.
3. The chipset transfers Spotify files from the SSD/HDD to RAM.
4. The CPU processes the instructions needed to start Spotify.
5. The chipset manages communication between the CPU, RAM, storage, and network adapter.
6. Spotify connects to the internet to access music content.
7. The CPU processes user actions such as searching and playing songs.
8. The chipset sends audio data to the sound system.
9. Music is played through the speakers or headphones.

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.



Ports

1. **USB 3.0 Type-A Port** – Used to connect USB devices such as keyboards, mice, and flash drives.
2. **SATA Ports** – Used to connect HDDs and SSDs.
3. **8-Pin CPU Power Connector** – Supplies power to the CPU.
4. **24-Pin ATX Power Connector** – Main power connection for the motherboard.
5. **Front Panel Header** – Connects the power button, reset button, and LED indicators.

Slots

1. **CPU Socket** – Holds the processor (CPU).
2. **RAM Slots (DIMM Slots)** – Used to install memory modules (RAM).
3. **PCIe x16 Slot** – Used for installing a graphics card (GPU).
4. **PCIe x1 Slot** – Used for expansion cards such as Wi-Fi cards and sound cards.
5. **M.2 Slot** – Used for installing high-speed NVMe SSDs.